12/08/19. UNIT-2.

Set Theory and Functions.

Set:

Set is a collection of elements which maintains some common property.

- Sets will be denoted by capital letters such as A, B, C, - - - -, X, Y, Z.
- Elements will be denoted by lower case letters such as a, b, c, - - -, x, y, z.
- The phrase "is belongs to" will be denoted by the symbol "$\in$".

Eg: $x \in A$

x is belongs to A.

$x \notin A$

x is not belongs to A.

Representation of sets:

1. Tabulation form.
2. Rules form.

Eg: $A = \{1, 2, 3, - - -\}$

$B = \{a, e, i, o, u\}$

Eg: $\{x / x \in N\}$

$\{x / x$ is a vowel$\}$

Types of sets:

1. Null set.
2. Singleton set.
3. Subset.
4. Universal set.
5. Complement set.

1. Null set:

A set is said to be null set if it doesn't contains any elements.

Eg: $\{\phi\}$ is called null set.

It is denoted as $\{\phi\}$.

2. Singleton set:

A set is said to be singleton set if it contains only one element.

Eg: $\{a\}$

$\{1\}$

3. Subset:

let A and B are two sets. then A is said to be subset of B if every element of 'A' is an element of 'B.'

Eg: $B = \{2, 3, 4, 6, 7\}$

$A = \{3, 6\}$

$\therefore A \subseteq B$.

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4. **Universal set:**

If all possible sets are subset of a fixed set then a fixed set is called as Universal set

Ex: It is denoted by symbol 'u'

5. **Complement set:**

A set contains all the elements of universal set except set A is called complement of A.

It can be denoted by 'A^c'.

Eg:

$u = \{1,2,3,4,5,6,7,8,9,10\}$.

$A = \{4,6,7,8\}$.

$A^c = u - A$

$A^c = \{1,2,3,5,9,10\}$.

Relations:

Relation of set A is R. which is defined as subset of $A \times A$.

i.e, $R \subseteq A \times A$.

(or).

Relation of sets A and B is R. which is defined as subset of $A \times B$.

i.e, $R \subseteq A \times B$.

Eg:

$A = \{1,2,3\}$ $B = \{1,2,3\}$. $R = \{(1,1)(1,2)(1,3)(2,3)\}$

$A \times B = \{(1,1)(1,2)(1,3)(2,1)(2,2)(2,3)(3,1)$
$(3,2)(3,3)\}$

$R \subseteq A \times B$.

Relation in matrix form: (M(R)).

let $A = \{1,2,3\}$, $B = \{1,2,3\}$ and

Relation is $R = \{(1,1)(2,2)(3,3)\}$.

To represent the relation R in matrix form we need to follow these steps

Step-1:

we need to find out $A \times B$.

$A \times B = \{(1,1)(1,2)(1,3)(2,1)(2,2)(2,3)(3,1)$
$(3,2)(3,3)\}$

Step-2:

The matrix form of $A \times B$.

$$M(A \times B) = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}_{3 \times 3}$$

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4. Universal set:

If all possible sets are subset of a fixed set then a fixed set is called as Universal set

It is denoted by symbol 'U'

5. Complement set:

A set contains all the elements of universal set except set A is called complement of A.

It can be denoted by A^c.

Eg:

$U = \{1,2,3,4,5,6,7,8,9,10\}$.

$A = \{4,6,7,8\}$.

$A^c = U - A$

$A^c = \{1,2,3,5,9,10\}$.

Relations:

Relation of set A is R. which is defined as subset of $A \times A$.

i.e, $R \subseteq A \times A$.

(or).

Relation of sets A and B is R. which is defined as subset of $A \times B$.

i.e, $R \subseteq A \times B$.

Eg:

$A = \{1,2,3\}$ $B = \{1,2,3\}$. $R = \{(1,1)(1,2)(1,3)(2,3)\}$

$A \times B = \{(1,1)(1,2)(1,3)(2,1)(2,2)(2,3)(3,1)$
$(3,2)(3,3)\}$

$R \subseteq A \times B$.

Relation in matrix form: (M(R)).

let $A = \{1,2,3\}$, $B = \{1,2,3\}$ and

Relation is $R = \{(1,1)(2,2)(3,3)\}$.

To represent the relation R in matrix form we need to follow these steps.

Step-1:

we need to find out $A \times B$.

$A \times B = \{(1,1)(1,2)(1,3)(2,1)(2,2)(2,3)(3,1)$
$(3,2)(3,3)\}$

Step-2:

The matrix form of $A \times B$.

$$M(A \times B) = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}_{3 \times 3}$$

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Step-3:

If $(a,b) \in A \times B$ and $(a,b) \in R$ then Result is '1' else result is '0'.

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

$(1,1) \in A \times B$ & $(1,1) \in R = 1$

$(1,2) \in A \times B$ & $(1,2) \notin R = 0$.

$(1,3) \in A \times B$ & $(1,3) \notin R = 0$.

$(2,1) \in A \times B$ & $(2,1) \notin R = 0$.

$(2,2) \in A \times B$ & $(2,2) \in R = 1$

$(2,3) \in A \times B$ & $(2,3) \notin R = 0$.

$(3,1) \in A \times B$ & $(3,1) \notin R = 0$.

$(3,2) \in A \times B$ & $(3,2) \notin R = 0$.

$(3,3) \in A \times B$ & $(3,3) \in R = 1$.

Step-4:

Represent the relation R in matrix form

$$M(R) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{3\times 3}$$

Step-5:

Complement of Relation R is R^C.

$$R^C = A \times B - R.$$

$$R^C = \{(1,2)\ (1,3)\ (2,1)\ (2,3)\ (3,1)\ (3,2)\}$$

$$M(R^C) = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}.$$

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R in matrix form:

Eg:

let $A = \{a_1, a_2, a_3\}$ and $B = \{b_1, b_2, b_3, b_4\}$.

$$R = \{(a_1,b_1)\ (a_1,b_4)\ (a_2,b_2)\ (a_2,b_3)\ (a_3,b_1)\ (a_3,b_3)\}.$$

Find the matrix form of R.

Sol:

$$A \times B = \{(a_1,b_1)\ (a_1,b_2)\ (a_1,b_3)\ (a_1,b_4)\ (a_2,b_1)\ (a_2,b_2)\ (a_2,b_3)\ (a_2,b_4)\ (a_3,b_1)\ (a_3,b_2)\ (a_3,b_3)\ (a_3,b_4)\}.$$

The matrix form of $A \times B$.

$$M(A \times B) = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{bmatrix}_{3\times 4}$$

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$(a_1, b_1) \in A \times B$ and $(a_1, b_1) \in R = 1$.

$(a_1, b_2) \in A \times B$ and $(a_1, b_2) \notin R = 0$.

⋮

$(a_3, b_4) \in A \times B$ and $(a_3, b_4) \notin R = 0$.

$$M(R) = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \end{bmatrix}_{3 \times 4}$$

Relation Representation by using graph:

A relation can be represented pictorially by drawing its graph as follows.

* A small circle is drawn for each element of A and marked with the corresponding elements. These circles called as "vertices".
* An arrow is drawn from the vertex a_i to the vertex "a_j" if and only if "$(a_i, a_j) \in R$". This is called "edge".
* Note that if $(a, a) \in R$ then arrow is drawn from a to a such edge is called as loop.

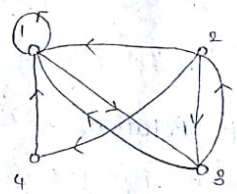
* This pictorial representation of R is called as Directed graph or Digraph.

Example:

1. Draw the directed graph of relation $R = \{(1,1)(1,3)(2,1)(2,3)(2,4)(3,1)(3,2)(4,1)\}$ on the set $\{1,2,3,4\}$.

Sol. The given relation is

$R = \{(1,1)(1,3)(2,1)(2,3)(2,4)(3,1)(3,2)(4,1)\}$

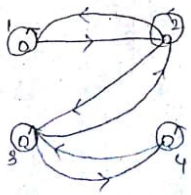


This is the directed graph of Relation R.

2. Draw the directed graph of relation $R = \{(1,1)(2,2)(3,3)(4,4)(1,2)(2,1)(2,3)(3,2)(4,3)(3,4)\}$ on the set $\{1,2,3,4\}$.

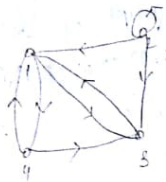
Sol. The given relation is

$R = \{(1,1)(2,2)(3,3)(4,4)(1,2)(2,1)(2,3)(3,2)(4,3)(3,4)\}$.



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3. What are the ordered pairs in relation R represented by the directed graph shown below?



Sol. The ordered pairs of Relation

$$R = \{(1,3), (1,4), (2,1)(2,2)(2,3), (3,1)(4,1)(4,3)\}$$

Representation of Relation:

This can be done by two ways.

1. Matrix form (Zero-one).
2. Directed Graph

Properties of Relation:

(Types of Relation):

These are 6 properties of Relation

Relation have following properties.

1. Reflexive.
2. Irreflexive.
3. Symmetric.
4. Asymmetric.
5. Anti-Symmetric.
6. Transitive.

*1. Reflexive:

If $a \in A$ & $(a,a) \in R$. then R is Reflexive.

Example:

let $A = \{1,2,3\}$

$R = \{(1,1)(2,2)(3,3)\}$.

$1 \in A$ & $(1,1) \in R$.

$2 \in A$ & $(2,2) \in R$.

$3 \in A$ & $(3,3) \in R$.

$\therefore$ R is Reflexive.

$A = \{1,2,3,4\}$

$R = \{(1,1)(2,2)(3,3)(4,4)$

$1 \in A$ & $(1,1) \in R$

$2 \in A$ & $(2,2) \in R$

$3 \in A$ & $(3,3) \in R$

$4 \in A$ & $(4,4) \notin R$

R is not Reflexive

2. Irreflexive:

If $a \in A$ & $(a,a) \notin R$ then R is Irreflexive.

Example:

let $A = \{a,b,c\}$.

$R = \{(a,b)(b,c)(c,c)(a,c)\}$.

$a \in A$ & $(a,a) \notin R$.

$b \in A$ & $(b,b) \notin R$.

$c \in A$ & $(c,c) \notin R$.

$\therefore$ R is not a Irreflexive.

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let $A = \{a, b, c\}$

$R = \{(a,b)(b,c)(c,a)(a,c)\}$

$a \in A$ & $(a,a) \notin R$.

$b \in A$ & $(b,b) \notin R$.

$c \in A$ & $(c,c) \notin R$.

$\therefore R$ is Irreflexive.

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* 3. Symmetric:

If $a, b \in A$ and $(a,b) \in R$
then $(b,a) \in R$
$\therefore R$ is Symmetric.

Eg:

$A = \{1, 2, 3, 4\}$

$R = \{(1,1)(1,2)(2,1)(1,3)(3,1)(2,3)(3,2)(4,2)(2,4)\}$

$\therefore R$ is

$(1,1) \in R$ & $(1,1) \in R$

$(1,2) \in R$ & $(2,1) \in R$.

$(1,3) \in R$ & $(3,1) \in R$.

$(2,3) \in R$ & $(3,2) \in R$.

$(4,2) \in R$ & $(2,4) \in R$.

$\therefore R$ is symmetric.

4. Asymmetric:

If $(a,b) \in A$ and $(a,b) \in R$
then $(b,a) \notin R$.
$\therefore R$ is Asymmetric.

Eg:

$A = \{1, 2, 3, 4\}$

$R = \{(1,2)(3,1)(2,4)(3,2)\}$

$(1,2) \in R$ & $(2,1) \notin R$.

$(3,1) \in R$ & $(1,3) \notin R$.

$(2,4) \in R$ & $(4,2) \notin R$.

$(3,2) \in R$ & $(2,3) \notin R$.

$\therefore R$ is Asymmetric.

5. Anti-Symmetric:

If $a, b \in A$ and $(a,b) \in R$
then $(b,a) \in R$ and $a = b$.
R is Anti-symmetric

Eg: $A = \{1, 2, 3\}$

$R = \{(1,1)(2,2)(3,3)\}$

$(1,1) \in R$ & $(1,1) \in R$.

$(2,2) \in R$ & $(2,2) \in R$.

$(3,3) \in R$ & $(3,3) \in R$.

$\therefore R$ is Anti-Symmetric.

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* 6. Transitive:

If $a, b, c \in A$ and $(a,b) \in R$ & $(b,c) \in R$ then $(a,c) \in R$.

∴ R is transitive.

Eg:

$A = \{1, 2, 3, 4\}$

$R = \{(1,2)(2,4)(1,4)(3,2)(2,1)(3,1)\}$

$(1,2) \in R$ and $(2,4) \in R$ then $(1,4) \in R$.

$(3,2) \in R$ and $(2,1) \in R$ then $(3,1) \in R$.

∴ R is Transitive.

* Equivalence Relation:

A relation R is said to be equivalence relation if it is reflexive, symmetric and transitive.

Eg:

1. Find whether R is equivalence relation or not. $A = \{1, 2, 3\}$ and $R = \{(1,1)(2,2)(3,3)(2,3)(3,1)\}$

Sol: Reflexive:

$1 \in A$ & $(1,1) \in R$.

$2 \in A$ & $(2,2) \in R$.

$3 \in A$ & $(3,3) \in R$.

∴ R is Reflexive.

Symmetric:

$(1,1) \in R$ & $(1,1) \in R$.

$(2,2) \in R$ & $(2,2) \in R$.

$(3,3) \in R$ & $(3,3) \in R$.

& R

$(2,3) \in R$ & $(3,2) \notin R$.

∴ R is not a symmetric.

∴ R is not an equivalence Relation.

2. Find whether R is equivalence relation or not. $A = \{1, 2, 3, 4\}$ and $R = \{(1,1)(2,2)(3,3)(4,4)(1,2)(2,1)(2,3)(3,2)(4,3)(3,4)\}$

Sol: Reflexive:

$1 \in A$ & $(1,1) \in R$.

$2 \in A$ & $(2,2) \in R$.

$3 \in A$ & $(3,3) \in R$.

$4 \in A$ & $(4,4) \in R$.

∴ R is Reflexive.

Symmetric:

$(1,1) \in R$ & $(1,1) \in R$.

$(2,2) \in R$ & $(2,2) \in R$.

$(3,3) \in R$ & $(3,3) \in R$.

$(4,4) \in R$ & $(4,4) \in R$.

$(1,2) \in R$ & $(2,1) \in R$.

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$(2,3) \in R$ & $(3,2) \in R$

$(4,3) \in R$ & $(3,4) \in R$.

$\therefore$ R is symmetric.

Transitive:

~~$(1,2) \in R$ and $(2,4) \in R$ then $(1,4)$~~

$(1,1) \in R$ and $(1,2) \in R$ then $(1,2) \in R$

$(1,2) \in R$ and $(2,1) \in R$ then $(1,1) \in R$.

$(2,3) \in R$ and $(3,2) \in R$ then $(2,2) \in R$.

$(4,3) \in R$ and $(3,4) \in R$ then $(4,4) \in R$

$\therefore$ R is Transitive.

$\therefore$ R is an Equivalence relation.

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Basic operations on matrices of Relations:

1. The relation matrix $M_{\tilde{R}}$ of $\tilde{R}$ can be obtained by simply interchanging the rows and columns of M_R. Such matrix is called as transpose of M_R

$$\therefore M_{\tilde{R}} = (M_R)^T$$

Eg:

The given relation matrix M_R, find $M_{\tilde{R}}$.

$$M_R = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

$$M_{\tilde{R}} = (M_R)^T$$

$$M_{\tilde{R}} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{\tilde{R}} = (M_R)^T$$

2. Boolean arithematic Operations:

These are two types of operations. Those are

- Join (∨)
- Meet (∧)

Join (∨):

$$b_1 \vee b_2 = \begin{cases} 1 & \text{if } b_1 = 1 \text{ or } b_2 = 1 \\ 0 & \text{otherwise.} \end{cases}$$

$1 \vee 1 = 1$
$1 \vee 0 = 1$
$0 \vee 1 = 1$
$0 \vee 0 = 0$

Meet (∧):

$$b_1 \wedge b_2 \begin{cases} 1 & \text{if } b_1 = 1 \text{ and } b_2 = 1 \\ 0 & \text{otherwise} \end{cases}$$

$1 \wedge 1 = 1$
$1 \wedge 0 = 0$
$0 \wedge 1 = 0$
$0 \wedge 0 = 0$

Eg:

1. Find the join and meet of the following matrix relations

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$M_R = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$ and $M_S = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$

Sol: Join:

$M_R \vee M_S$

$= \begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \vee \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$

$= \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$

$= M_{R \cup S}$

Meet:

$M_R \wedge M_S$

$= \begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \wedge \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$

$= \begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

$= M_{R \cap S}$

3. Boolean product:

The Boolean product of A and B, denoted by $A \odot B$.

Eg:

1. Find boolean product of the following matrices relations.

$M_R = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}_{3 \times 2}$ $M_S = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}_{2 \times 3}$

Sol: The boolean product of M_R and M_S is

$M_R \odot M_S = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} \odot \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$

$= \begin{bmatrix} (1 \wedge 1) \vee (0 \wedge 0) & (1 \wedge 1) \vee (0 \wedge 1) & (1 \wedge 0) \vee (0 \wedge 1) \\ (0 \wedge 1) \vee (1 \wedge 0) & (0 \wedge 1) \vee (1 \wedge 1) & (0 \wedge 0) \vee (1 \wedge 1) \\ (1 \wedge 1) \vee (1 \wedge 0) & (1 \wedge 1) \vee (0 \wedge 1) & (1 \wedge 0) \vee (0 \wedge 1) \end{bmatrix}$

$= \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$

$= M_{S \circ R}$

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Q. Find the boolean product of the following matrices relations.

$$M_R = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}_{3\times 3} \quad M_S = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}_{3\times 3}$$

Sol. The boolean product of M_R and M_S is $M_R \odot M_S$

$$= \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \odot \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} (1\wedge 0)\vee(0\wedge 0)\vee(1\wedge 1) & (1\wedge 1)\vee(0\wedge 0)\vee(1\wedge 0) & (1\wedge 0)\vee(0\wedge 1)\vee(1\wedge 1) \\ (1\wedge 0)\vee(1\wedge 0)\vee(0\wedge 1) & (1\wedge 1)\vee(1\wedge 0)\vee(0\wedge 0) & (1\wedge 0)\vee(1\wedge 1)\vee(0\wedge 1) \\ (0\wedge 0)\vee(0\wedge 0)\vee(0\wedge 1) & (0\wedge 1)\vee(0\wedge 1)\vee(0\wedge 0) & (0\wedge 0)\vee(0\wedge 1)\vee(0\wedge 1) \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$M_{ROS} = M_{SOR}$.

** Transitive closure (R^+):

Transitive closure of relation 'R' on set 'A' is the smallest transitive relation on 'A' containing 'R'.

∴ If set 'A' containing n elements, then the transitive closure of 'R' on set 'A' is

$$R^+ = R \cup R^2 \cup R^3 \cup \cdots R^n.$$

(OR)

$$M_R{}^+ = M_R \vee M_R^{[2]} \vee M_R^{[3]} \vee \cdots M_R^{[n]}.$$

Eg:

1. Find the matrix of transitive closure of 'R' whose $M_R = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$

Sol. The transitive closure of M_R is $M_R{}^+$

$$M_R{}^+ = M_R \vee M_R^{[2]} \vee M_R^{[3]}$$

$$M_R^{[2]} = M_R \odot M_R$$

$$= \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix} \odot \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

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$$M_R^{[3]} = M_R^{[2]} \odot M_R$$

$$= \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \odot \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

$$M_R^+ = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix} \vee \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \vee \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

2. Find the matrix of transitive closure of 'R' where $A = \{1, 2, 3\}$ and $R = \{(1,2)(2,3)(3,1)(3,2)\}$.

Sol:

The given relation $R = \{(1,2)(2,3)(3,1)(3,2)\}$

Matrix of Relation $M_R = \begin{array}{c} \\ 1 \\ 2 \\ 3 \end{array} \begin{array}{c} \begin{array}{ccc} 1 & 2 & 3 \end{array} \\ \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \end{array}$

The transitive closure of M_R is M_R^+

$$M_R^+ = M_R \vee M_R^{[2]} \vee M_R^{[3]}$$

$$M_R^{[2]} = M_R \odot M_R$$

$$= \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \odot \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

$$M_R^{[3]} = M_R^{[2]} \odot M_R$$

$$= \begin{bmatrix} 0 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \odot \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$M_R^+ = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \vee \begin{bmatrix} 0 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \vee \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

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23-08-14.

** **Warshall's Algorithm:**

This is an efficient method for computing transitive closure of a relation.

Algorithm:

The given matrix relation is M_R of a simple directed graph and compute M_R^+ (transitive closure) as follows.

step-1: $P^{[0]} = M_R$

step-2: $k = 1$

step-3: $i = 1$

step-4: $P_{ij}^{[k]} = P_{ij}^{[k-1]} \vee \left(P_{ik}^{[k-1]} \wedge P_{kj}^{[k-1]} \right)$

$\forall j = 1$ to n.

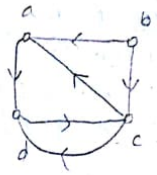
step-5: $i = i+1$ if $i \le n$, goto step 4.

step-6: $k = k+1$ if $k \le n$, goto step 3.

Otherwise stop.

Example:

1. Let R be a relation with directed graph. Find the matrix of relation (M_R) and transitive closure of matrix relation (M_R^+) by using Warshall's Algorithm.



Sol: Matrix of relation (M_R) =

	a	b	c	d
a	0	0	0	1
b	1	0	1	0
c	1	0	0	1
d	0	0	1	0

M_R = $P^{[0]}$

	a	b	c	d
a	0	0	0	1
b	1	0	1	0
c	1	0	0	1
d	0	0	1	0

b-a | a-d
c-a

b-d = 1
c-d = 1

$P^{[1]}$ =

	a	b	c	d
a	0	0	0	1
b	1	0	1	1
c	1	0	0	1
d	0	0	1	0

$P^{[2]}$ =

	a	b	c	d
a	0	0	0	1
b	1	0	1	1
c	1	0	0	1
d	1	0	1	1

b-c | c-a
d-c | c-d

b-a
b-d
d-a
d-d

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$P^{[3]} =$

	a	b	c	d
a	0	0	0	1
b	1	0	1	1
c	1	0	0	1
d	1	0	1	1

a-d, b-d, c-d, d-d | d-a, d-c, d-d

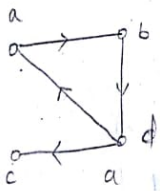
$P^{[4]} =$

	a	b	c	d
a	1	0	1	1
b	1	0	1	1
c	1	0	1	1
d	1	0	1	1

a-c, a-d, b-c, b-d, c-c, c-d, d-c, d-d | a-a, a-a, c-a, d-a

This is the transitive closure of MR.

2.



Sol.

$M_R =$

	a	b	c	d
a	0	1	0	0
b	0	0	0	1
c	0	0	0	0
d	1	0	1	0

$M_R =$

	a	b	c	d
a	0	1	0	0
b	0	0	0	1
c	0	0	0	0
d	1	0	1	0

$= P^{[0]}$

d-a | a-b

d-b = 1

$P^{[1]} =$

	a	b	c	d
a	0	1	0	0
b	0	0	0	1
c	0	0	0	0
d	1	1	1	0

a-b, d-b | b-d

a-d, d-d

$P^{[2]} =$

	a	b	c	d
a	0	1	0	1
b	0	0	0	1
c	0	0	0	0
d	1	1	1	1

$P^{[3]} =$

	a	b	c	d
a	0	1	0	1
b	0	0	0	1
c	0	0	0	0
d	1	1	1	1

a-d, b-d, d-d | d-a, d-b, d-c, d-d

a-a b-a d-a
a-b b-b d-b
a-c b-c d-c
a-d b-d d-d

$P^{[4]} =$

	a	b	c	d
a	1	1	1	1
b	1	1	1	1
c	0	0	0	0
d	1	1	1	1

This is the transitive closure of M_R.

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25-08-14.

Compatability Relation:

A relation R is said to be compatability relation if it is reflexive and symmetric

Eg:

Reflexive: If $a \in A$, $(a,a) \in R$.

Symmetric: If $a, b \in A$, $(a,b) \in R$ & $(b,a) \in R$.

Eg:

1. $A = \{1,2,3,4\}$ and $R = \{(1,1)(2,3)(4,3)(3,2)(2,2)(3,4)(3,3)(4,4)\}$ Find whether R is a compatability relation or not.

Sol.

Reflexive:

$1 \in A$ & $(1,1) \in R$.

$2 \in A$ & $(2,2) \in R$.

$3 \in A$ & $(3,3) \in R$.

$4 \in A$ & $(4,4) \in R$.

$\therefore$ R is Reflexive.

Symmetric:

$(1,1) \in R$ & $(1,1) \in R$.

$(2,3) \in R$ & $(3,2) \in R$.

$(4,3) \in R$ & $(3,4) \in R$.

$\therefore$ R is symmetric.

$\therefore$ R is Compatability relation.

2. $A = \{1,2,3,4\}$, (i) $R_1 = \{(1,2)(2,3)(3,4)(4,1)(1,1)(2,2)(3,3)(4,4)\}$, (ii) $R_2 = \{(3,2)(2,3)(1,1)(2,2)(3,3)(4,4)\}$. find whether given relations are compatability relations or not.

Sol. (i)

Reflexive:

$1 \in A$ & $(1,1) \in R_1$

$2 \in A$ & $(2,2) \in R_1$

$3 \in A$ & $(3,3) \in R_1$

$4 \in A$ & $(4,4) \in R_1$

$\therefore$ R_1 is Reflexive.

Symmetric:

$(1,1) \in R_1$ & $(1,1) \in R_1$

$(1,2) \in R_1$ & $(2,1) \notin R_1$

$\therefore$ R_1 is not symmetric

$\therefore$ R_1 is not a compatability relation.

(ii)

Reflexive:

$1 \in A$ & $(1,1) \in R_2$

$2 \in A$ & $(2,2) \in R_2$

$3 \in A$ & $(3,3) \in R_2$

$4 \in A$ & $(4,4) \in R_2$

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∴ R_2 is Reflexive.

Symmetric:

$(3,2) \in R_2$ & $(2,3) \in R_2$

$(1,1) \in R_2$ & $(1,1) \in R_2$

∴ R_2 is symmetric.

∴ R_2 is compatability relation.

Partial Order Set: (POSET)

A relation R is said to be partial order set if it is reflexive, transitive and anti-symmetric.

Reflexive: $a \in A$ & $(a,a) \in R$.

Transitive: $(a,b) \in R$ & $(b,c) \in R$ then $(a,c) \in R$.

Anti-Symmetric: $(a,b) \in R$ & $(b,a) \in R$ then $a = b$.

Eg:

1. Find $A = \{1,2,3,4\}$, $R = \{(1,2)(2,3)(1,1)(1,3)(2,2)(3,3)(4,4)\}$ Find whether R is a poset or not.

Sol.

Reflexive:

$1 \in A$ & $(1,1) \in R$.

$2 \in A$ & $(2,2) \in R$

$3 \in A$ & $(3,3) \in R$

$4 \in A$ & $(4,4) \in R$.

∴ R is Reflexive.

Transitive:

$(1,2) \in R$ & $(2,3) \in R$ then $(1,3) \in R$.

∴ R is transitive.

Anti-Symmetric:

$(1,1) \in R$ & $(1,1) \in R$ then $1 = 1$

$(2,2) \in R$ & $(2,2) \in R$ then $2 = 2$

$(3,3) \in R$ & $(3,3) \in R$ then $3 = 3$.

$(4,4) \in R$ & $(4,4) \in R$ then $4 = 4$

∴ R is Anti-symmetric.

∴ R is a poset.

2. $A = \{1,2,3,4\}$, & $R = \{(3,2)(2,3)(1,1)(2,2)(3,3)(4,4)\}$ Find whether R is a poset or not.

Sol.

Reflexive:

$1 \in A$ & $(1,1) \in R$.

$2 \in A$ & $(2,2) \in R$.

$3 \in A$ & $(3,3) \in R$

$4 \in A$ & $(4,4) \in R$.

∴ R is Reflexive.

Transitive:

$(3,2) \in R$ & $(2,3) \in R$ then $(3,3) \in R$.

∴ R is transitive.

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Anti-Symmetric:

$(1,1) \in R$ & $(1,1) \in R$ then $1=1$

$(2,2) \in R$ & $(2,2) \in R$ then $1=1$

$(3,3) \in R$ & $(3,3) \in R$ then $3=3$

$(4,4) \in R$ & $(4,4) \in R$ then $4=4$

$(3,2) \in R$ & $(2,3) \in R$, then $3 \neq 2$

$\therefore$ R is not a Anti-Symmetric

$\therefore$ R is not a Poset

* Hasse Diagram:

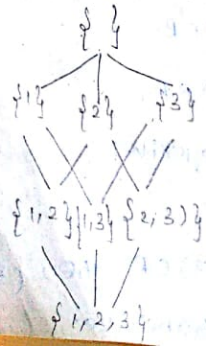
A pictorial representation of a relation R and all subsets relations of R.

Eg:

1. Find Hasse diagram of relation $R=\{1,2,3\}$

Sol: Subsets of relation R.

$P = \{\{\}, \{1\}, \{2\}, \{3\}, \{1,2\}, \{1,3\}, \{2,3\}, \{1,2,3\}\}$



2. Find Hasse diagram of relation $R=\{a,b,c,d\}$

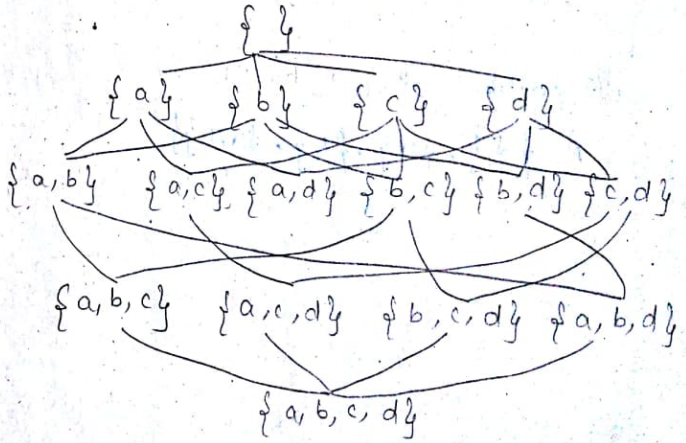
Sol: Subsets of relation R.

$P = \{\{\}, \{a\}, \{b\}, \{c\}, \{d\}, \{a,b\}, \{b,c\}, \{a,c\}, \{a,d\}, \{b,d\}, \{c,d\}, \{a,b,c\}, \{a,c,d\}, \{b,c,d\}, \{a,b,d\}, \{a,b,c,d\}\}$

{ }

{a} {b} {c} {d}

{a,c} {a,b} {b,c} {a,d}



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Q6 (a)

1. Draw the Hasse diagram of the following

(i) $\{a\}$

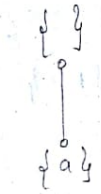
(ii) $\{a, b\}$

(iii) $\{a, b, c\}$

(iv) $\{a, b, c, d\}$

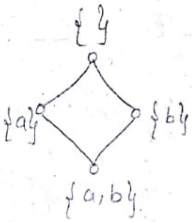
(i) $\{a\}$ $\quad 2^1 = 2$,

$P = \{\{\}, \{a\}\}$



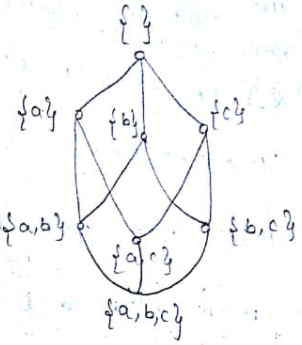
(ii) $\{a, b\}$ $\quad 2^2 = 4$

$P = \{\{\}, \{a\}, \{b\}, \{a, b\}\}$



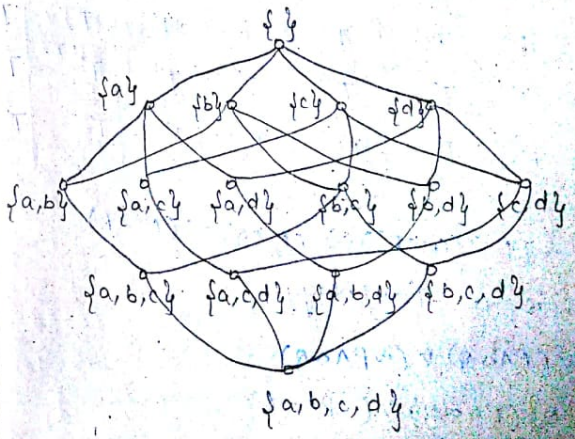
(iii) $\{a, b, c\}$ $\quad 2^3 = 8$.

$P = \{\{\}, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\}$



(iv) $\{a, b, c, d\}$ $\quad 2^4 = 16$.

$P = \{\{\}, \{a\}, \{b\}, \{c\}, \{d\}, \{a, b\}, \{a, c\}, \{a, d\}, \{b, c\}, \{b, d\}, \{c, d\}, \{a, b, c\}, \{a, c, d\}, \{a, b, d\}, \{a, d, b\}, \{b, c, d\}, \{a, b, c, d\}\}$



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1. Specify that the following statements are tautology or not.

(i) $(P \to (Q \to R)) \Leftrightarrow (P \to Q) \to (P \to R)$

Sol. $(P \to (Q \to R)) \Leftrightarrow (P \to Q) \to (P \to R)$

P Q R $P \to Q$ $P \to R$ $Q \to R$ $P \to a$

P	Q	R	$P \to Q$ a	$P \to R$ b	$Q \to R$ c	$a \to b$ e	$P \to c$ d	d⇔e
T	T	T	T	T	T	T	T	T
T	T	F	T	F	F	F	F	T
T	F	T	F	T	T	T	T	T
T	F	F	F	F	T	T	T	T
F	T	T	T	T	T	T	T	T
F	T	F	T	T	F	T	T	T
F	F	T	T	T	T	T	T	T
F	F	F	T	T	T	T	T	T

$\therefore (P \to (Q \to R)) \Leftrightarrow (P \to Q) \to (P \to R)$ is a Tautology.

(ii) $Q \vee (P \wedge \sim Q) \vee (\sim P \wedge \sim Q)$

Sol. $Q \vee (P \wedge \sim Q) \vee (\sim P \wedge \sim Q)$

P	Q	$\sim P$	$\sim Q$	$P \wedge \sim Q$ a	$\sim P \wedge \sim Q$ b	$a \vee b$ c	$Q \vee c$
T	T	F	F	F	F	F	T
T	F	F	T	T	F	T	T
F	T	T	F	F	F	F	T
F	F	T	T	F	T	T	T

$\therefore Q \vee (P \wedge \sim Q) \vee (\sim P \wedge \sim Q)$ is a Tautology.

2. Prove that the statement $\neg(P \rightleftarrows Q) \Leftrightarrow (P \vee Q) \wedge \neg(P \wedge Q)$ is an equivalence formula by using truth table method & replacement method.

Sol. $\neg(P \rightleftarrows Q) \Leftrightarrow (P \vee Q) \wedge \neg(P \wedge Q)$

Truth table Method.

P	Q	$P \vee Q$ a	$P \wedge Q$	$\neg(P \wedge Q)$ b	$a \wedge b$ d	$P \rightleftarrows Q$ c	1c	1c⇔d
T	T	T	T	F	F	T	F	T
T	F	T	F	T	T	F	T	T
F	T	T	F	T	T	F	T	T
F	F	F	F	T	F	T	F	T

$\neg(P \rightleftarrows Q) \Leftrightarrow (P \vee Q) \wedge \neg(P \wedge Q)$ is a tautology. So it is Equivalence formula.

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Replacement Method:

$\neg(P \rightleftarrows Q) \Leftrightarrow (P \vee Q) \wedge \neg(P \wedge Q)$

The given statement formula is

$\neg(P \rightleftarrows Q) \Leftrightarrow (P \vee Q) \wedge \neg(P \wedge Q)$

consider L.H.S.

$\neg(P \rightleftarrows Q)$ $\quad [\because P \rightleftarrows Q = (P \rightarrow Q) \wedge (Q \rightarrow P)]$

$\Leftrightarrow \neg((P \rightarrow Q) \wedge (Q \rightarrow P))$ $\quad [\because P \rightarrow Q = \neg P \vee Q]$.

$\Leftrightarrow \neg(\neg P \vee Q) \wedge (\neg Q \vee P))$

$\Leftrightarrow (P \wedge \neg Q) \wedge \vee (Q \wedge \neg P)$ $\quad [\because (a \wedge b) \vee (c \wedge d) = (P \vee Q) \wedge (b \vee d)]$

~~$\Leftrightarrow (P \vee Q) \wedge \neg(Q \wedge \neg P)$~~

~~$\Leftrightarrow (P \vee Q) \wedge \neg(P \wedge Q)$~~ $\quad (a \wedge c) \vee (a \wedge d) \vee (b \wedge c) \vee (b \wedge d)$

$\Leftrightarrow (P \vee Q) \wedge (\neg Q \vee \neg P)$

$\Leftrightarrow (P \vee Q) \wedge \neg(Q \wedge P)$

$\Leftrightarrow (P \vee Q) \wedge \neg(P \wedge Q)$

which is R.H.S.

Hence the given formula is equivalent.

3. a) Obtain d.n.f of $P \rightarrow ((P \rightarrow Q) \wedge \neg(\neg Q \vee \neg P))$ without using truth tables.

Sol: The given formula.

$\Leftrightarrow P \rightarrow ((P \rightarrow Q) \wedge \neg(\neg Q \vee \neg P))$

$\Leftrightarrow P \rightarrow ((\neg P \vee Q) \wedge \neg(\neg Q \vee \neg P))$

$\Leftrightarrow P \rightarrow ((\neg P \vee Q) \wedge (Q \wedge P))$

$\Leftrightarrow \neg P \vee ((\neg P \vee Q) \wedge (Q \wedge P))$ (S)

$\Leftrightarrow \neg P \vee ((\neg P \wedge Q \wedge P) \vee (Q \wedge Q \wedge P))$

$\Leftrightarrow \neg P \vee (\neg P \wedge P) \wedge Q \vee (Q \wedge Q) \wedge P)$

$\Leftrightarrow \neg P \vee (F \wedge Q) \vee (Q \wedge P)$

$\Leftrightarrow \neg P \vee F \vee (P \wedge Q)$

$\Leftrightarrow \neg P \vee (P \wedge Q)$

which is dnf of given formula.

b) Obtain p.d.n.f of above statement using truth tables.

Sol:

P	Q	$\neg P$	$\neg Q$	$P \rightarrow Q$ (b)	$\neg Q \vee \neg P$ (a)	$\neg Q$	$b \wedge \neg a$ (c)	$p \rightarrow c$
T	T	F	F	T	F	T	T	T
T	F	F	T	F	T	F	F	F
F	T	T	F	T	T	F	F	T
F	F	T	T	T	T	F	F	T

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Minterms.
$P \wedge Q$
$P \wedge \neg Q$
$\neg P \wedge Q$
$\neg P \wedge \neg Q$

In the above example, the truth value "T" is in 1st, 3rd and 4th rows. then the assigned truth values are

P and Q are T, T ; F, T ; F, F ;

then the p.d.n.f of given formula is

$$(P \wedge Q) \vee (\neg P \wedge Q) \vee (\neg P \wedge \neg Q)$$

4. let $P(x)$: x is a professional athelete,

$Q(x)$: x plays soccer.

Write the following into English & write the negation.

(i) $\forall x\,(P(x) \rightarrow Q(x))$.

Sol: For all x, if x is a professional athlete then x plays soccer.

Negation:

$$\neg(\forall x\,(P(x) \rightarrow Q(x)))$$
$$= \exists x\ \neg(P(x) \rightarrow Q(x))$$
$$= \exists x\ \neg(\neg(P(x)) \vee Q(x))$$
$$= \exists x\ (P(x) \wedge \neg Q(x))$$

for some x, x is a professional athlete and x does not plays soccer.

(ii) $\exists x\,(P(x) \wedge Q(x))$.

Sol: for some x, x is a professional athlete and x plays soccer.

Negation:

$$\neg(\exists x\,(P(x) \wedge Q(x)))$$
$$= \forall x\ \neg(P(x) \wedge Q(x))$$
$$= \forall x\,(\neg P(x) \vee \neg Q(x))$$

For all x, x is not a professional athlete or x does not plays soccer.

(iii) $\exists x\,(P(x) \vee Q(x))$

Sol: for some x, x is a professional athlete or x plays soccer.

Negation:

$$\neg(\exists x\,(P(x) \vee Q(x)))$$
$$\forall x\ \neg(P(x) \vee Q(x))$$
$$\forall x\,(\neg P(x) \wedge \neg Q(x))$$

for all x, x is not a professional athlete and x does not plays soccer.

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5. (i) Show that $R \vee S$ can be valid conclusion from the given premises $C \vee D$, $(C \vee D) \rightarrow \neg H$, $\neg H \rightarrow (A \wedge \neg B)$ and $(A \wedge \neg B) \rightarrow (R \vee S)$.

Sol. The given premises are

$(C \vee D)$, $(C \vee D) \rightarrow \neg H$, $\neg H \rightarrow (A \wedge \neg B)$ and $(A \wedge \neg B) \rightarrow (R \vee S)$.

[1]	(1)	$C \vee D$	Rule P
[2]	(2)	$(C \vee D) \rightarrow \neg H$	Rule P.
[1, 2]	(3)	$\neg H$	Rule T (1), (2)
[4]	(4)	$\neg H \rightarrow (A \wedge \neg B)$	Rule P
[1, 2, 4]	(5)	$A \wedge \neg B$	Rule T (3), (4)
[6]	(6)	$(A \wedge \neg B) \rightarrow (R \vee S)$	Rule P.
[1, 2, 4, 6]	(7)	$R \vee S$	Rule T (5), (6)

$\therefore R \vee S$ is a valid conclusion.

(ii) Show that by direct proof

$R \rightarrow (S \rightarrow Q)$, $\neg P \vee R$, $S \Rightarrow P \rightarrow Q$

Sol. The given premises are

$R \rightarrow (S \rightarrow Q)$, $\neg P \vee R$, S

$P \rightarrow Q$

[1]	(1)	$R \rightarrow (S \rightarrow Q)$	Rule P.
[2]	(2)	$\neg P \vee R$	Rule P
[1, 2]	(3)	$\neg P \vee (S \rightarrow Q)$	Rule T (1), (2)
[4]	(4)	S	Rule P
[1, 2, 4]	(5)	$\neg P \vee Q$	Rule T (3), (4)
[6]	(6)	$P \rightarrow Q$	Rule T (5)

$\therefore P \rightarrow Q$ is a valid conclusion.

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6. Proof by indirect method.

(i) $P \to Q,\ Q \to R,\ \neg(P \wedge R),\ P \vee R \Rightarrow R$.

Sol. The desired conclusion 'R', '$\neg R$' is a new premise.

[1]	(1)	$P \vee R$	Rule P
[2]	(2)	$\neg R$	Rule P (additional premises).
[1,2]	(3)	P	Rule T (1),(2)
[4]	(4)	$P \to Q$	Rule P
[1,2,4]	(5)	Q	Rule T (3),(4)
[6]	(6)	$Q \to R$	Rule P
[1,2,4,6]	(7)	R	Rule T (5),(6)
[8]	(8)	$\neg(P \wedge R)$	Rule P
[9]	(9)	$\neg P \vee \neg R$	Rule T (8)
[1,2,4,6,9]	(10)	$\neg P$	Rule T (7),(9)
[1,2,4,6,9]	(11)	$P \wedge \neg P$	Rule T (3),(10)

$P \wedge \neg P$ is a contradiction.

Thus R is the valid conclusion of given premises.

(ii) $P \to Q,\ R \to S,\ P \vee R \Rightarrow Q \vee S$.

Sol. The desired conclusion '$Q \vee S$', '$\neg(Q \vee S)$' is a new premise.

[1]	(1)	$P \to Q$	Rule P
[2]	(2)	$\neg(Q \vee S)$	Rule P (additional premise)
[3]	(3)	$\neg Q \wedge \neg S$	Rule T (2)
[1,3]	(4)	$\neg P \wedge \neg S$	Rule T (1),(3)
[5]	(5)	$R \to S$	Rule P
[1,3,5]	(6)	$\neg P \wedge \neg R$	Rule T (4),(5)
[7]	(7)	$P \vee R$	Rule P
[1,3,5,7]	(8)	$\neg P \wedge P$	Rule T (6),(7)

$\neg P \wedge P$ is a contradiction.

Thus $Q \vee S$ is the valid conclusion of given premises.

7. $A = \{1,2,3,4\}$ & $R = \{(1,1)(2,2)(3,3)(4,4)(1,2)(2,1)(2,3)(3,2)(4,3)(3,4)\}$ find whether the relation R is (i) equivalence or not (ii) poset or not (iii) Compatability relation or not.

Sol (i) Equivalence

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Reflexive:

$1 \in A$ & $(1,1) \in R$.
$2 \in A$ & $(2,2) \in R$
$3 \in A$ & $(3,3) \in R$
$4 \in A$ & $(4,4) \in R$.
$\therefore$ R is Reflexive.

Symmetric:

$(1,1) \in R$ & $(1,1) \in R$.
$(2,2) \in R$ & $(2,2) \in R$.
$(3,3) \in R$ & $(3,3) \in R$.
$(4,4) \in R$ & $(4,4) \in R$.
$(1,2) \in R$ & $(2,1) \in R$.
$(2,3) \in R$ & $(3,2) \in R$
$(4,3) \in R$ & $(3,4) \in R$.
$\therefore$ R is Symmetric

Transitive:

$(1,1) \in R$ & $(1,2) \in R$ then $(1,2) \in R$.
$(1,2) \in R$ & $(2,1) \in R$ then $(1,1) \in R$
$(2,3) \in R$ & $(3,2) \in R$. then $(2,2) \in R$
$(4,3) \in R$ & $(3,4) \in R$ then $(4,4) \in R$.
$\therefore$ R is transitive.

$\therefore$ R is Equivalence relation.

(iii) POSET:

Reflexive:

$1 \in A$ & $(1,1) \in R$
$2 \in A$ & $(2,2) \in R$
$3 \in A$ & $(3,3) \in R$
$4 \in A$ & $(4,4) \in R$.
$\therefore$ R is Reflexive.

Transitive:

$(1,1) \in R$ & $(1,2) \in R$ then $(1,2) \in R$.
$(1,2) \in R$ & $(2,1) \in R$ then $(1,1) \in R$.
$(2,3) \in R$ & $(3,2) \in R$ then $(2,2) \in R$.
$(4,3) \in R$ & $(3,4) \in R$ then $(4,4) \in R$.
$\therefore$ R is Transitive.

Anti-Symmetric:

$(1,1) \in R$ & $(1,1) \in R$ then $1 = 1$
$(2,2) \in R$ & $(2,2) \in R$ then $2 = 2$
$(3,3) \in R$ & $(3,3) \in R$ then $3 = 3$
$(4,4) \in R$ & $(4,4) \in R$ then $4 = 4$.
$(1,2) \in R$ & $(2,1) \in R$ then $x \in R$ $1 \neq 2$
$(2,3) \in R$ & $(3,2) \in R$ then $2 \neq 3$
$(4,3) \in R$ & $(3,4) \in R$ then $4 \neq 3$.
$\therefore$ R is not a Anti-Symmetric

$\therefore$ R is not a POSET.

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(iii) Compatability relation:

As the relation is equivalence it is also a compatability relation.

Reflexive:

$1 \in A$ & $(1,1) \in R$.

$2 \in A$ & $(2,2) \in R$.

$3 \in A$ & $(3,3) \in R$.

$4 \in A$ & $(4,4) \in R$.

$\therefore R$ is Reflexive.

Symmetric:

$(1,1) \in R$ & $(1,1) \in R$.

$(2,2) \in R$ & $(2,2) \in R$.

$(3,3) \in R$ & $(3,3) \in R$.

$(4,4) \in R$ & $(4,4) \in R$.

$(1,2) \in R$ & $(2,1) \in R$.

$(2,3) \in R$ & $(3,2) \in R$.

$(4,3) \in R$ & $(3,4) \in R$.

$\therefore R$ is Symmetric.

$\therefore R$ is compatability relation.

8. Find the matrix of transitive closure of relation R where $A = \{a, b, c, d\}$ & $R = \{(a,d), (b,a), (b,c), (c,a), (c,d), (d,c)\}$

Sol: The given relation $R = \{(a,d)(b,a)(b,c)(c,a)(c,d)(d,c)\}$

Matrix of Relation $M_R =$

	a	b	c	d
a	0	0	0	1
b	1	0	1	0
c	1	0	0	1
d	0	0	1	0

The transitive closure of M_R is M_R^+

$$M_R^+ = M_R \vee M_R^{[2]} \vee M_R^{[3]} \vee M_R^{[4]}$$

$$M_R^{[2]} = M_R \odot M_R$$

$$= \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \odot \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \end{bmatrix}$$

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$M_R^{[3]} = M_R^{[2]} \odot M_R$

$$= \begin{bmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \end{bmatrix} \odot \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

$M_R^{[4]} = M_R^{[3]} \odot M_R$

$$= \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix} \odot \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

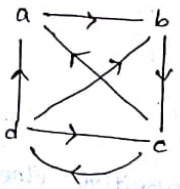
$$= \begin{bmatrix} 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix}$$

$M_R^+ = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \vee \begin{bmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \end{bmatrix} \vee \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix} \vee$

$$\begin{bmatrix} 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix}.$$

$$= \begin{bmatrix} 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix}.$$

q. let R be the relation with directed graph. Find the matrices. M_R & M_R^+ by Warshall's Algorithm.



Sol: Matrix of relation (M_R) =

	a	b	c	d
a	0	1	0	0
b	0	0	1	0
c	1	0	0	0
d	1	1	1	0

M_R = (same matrix) $= P^{[0]}$.

c-a | a-b
d-a |

c-b = 1
d-b = 1

$P^{[1]}$ =

	a	b	c	d
a	0	1	0	0
b	0	0	1	0
c	1	1	0	1
d	1	1	1	0

a-b | b-c.
c-b |
d-b |

a-c
c-c
d-c

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$P^{[2]} =$

	a	b	c	d
a	0	1	1	0
b	0	0	1	1
c	1	1	1	0
d	1	1	1	0

a-c	c-a
b-c	c-b
c-c	c-c
d-c	c-d

$P^{[3]} =$

	a	b	c	d
a	1	1	1	1
b	1	1	1	1
c	1	1	1	1
d	1	1	1	1

a-a	c-a
a-b	c-b
a-c	c-c
a-d	c-d
b-a	d-a
b-b	d-b
b-c	d-c
b-d	d-d

$= P^{[4]}$

This is the transitive closure of M_R.

10. Find the join, meet and product of M_R and M_S where $M_R = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$ $M_S = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$.

Join:

$M_R \vee M_S$

$= \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \vee \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$

$= \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

$= M_{R \cup S}$.

Meet:

$M_R \wedge M_S$

$= \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \wedge \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$

$= \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$

$= M_{R \cap S}$

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The product of M_R and M_S is

$$M_R \odot M_S = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \odot \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} (1\wedge1)\vee(0\wedge1)\vee(1\wedge0) & (1\wedge0)\vee(0\wedge0)\vee(1\wedge0) & (1\wedge0)\vee(0\wedge1)\vee(1\wedge0) \\ (1\wedge1)\vee(1\wedge1)\vee(0\wedge0) & (1\wedge0)\vee(1\wedge0)\vee(0\wedge1) & (1\wedge0)\vee(1\wedge1)\vee(0\wedge0) \\ (1\wedge1)\vee(1\wedge1)\vee(1\wedge0) & (1\wedge0)\vee(1\wedge0)\vee(1\wedge1) & (1\wedge0)\vee(1\wedge1)\vee(1\wedge0) \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$= M_{S \circ R}$

2. $\neg(P \rightleftarrows Q) \Leftrightarrow (P \vee Q) \wedge \neg(P \wedge Q)$

Replacement method:

Consider L.H.S

$\neg(P \rightleftarrows Q)$

$[\because P \rightleftarrows Q = (P \rightarrow Q) \wedge (Q \rightarrow P)]$

$\Leftrightarrow \neg[(P \rightarrow Q) \wedge (Q \rightarrow P)]$

$[\because P \rightarrow Q = \neg P \vee Q]$

$\Leftrightarrow \neg((\neg P \vee Q) \wedge (\neg Q \vee P))$

$\Leftrightarrow (P \wedge \neg Q) \vee (Q \wedge \neg P)$ $[\because (a\wedge b)\vee(c\wedge d) = (a\vee c)\wedge(a\vee d)\wedge(b\vee c)\wedge(b\vee d)]$

$\Leftrightarrow (P \vee Q) \wedge (P \vee \neg P) \wedge (\neg Q \vee Q) \wedge (\neg Q \vee \neg P)$

$\Leftrightarrow (P \vee Q) \wedge T \wedge T \wedge (\neg Q \vee \neg P)$ $[\because P \vee \neg P = T]$

$\Leftrightarrow (P \vee Q) \wedge (\neg Q \vee \neg P)$ $[\because P \wedge T = P]$

$\Leftrightarrow (P \vee Q) \wedge \neg(Q \wedge P)$

$\Leftrightarrow (P \vee Q) \wedge \neg(P \wedge Q)$

$=$ R.H.S.

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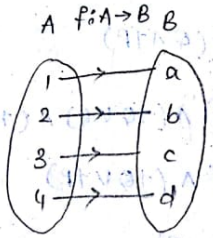
Functions:

Defination:

A mapping $f: A \to B$ is said to be function if every element of 'A' having unique image in 'B'.

if A, B are two sets, then A is domain and B is range

Representation of function:



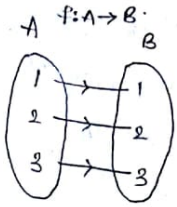
Types of functions:

1. Identity function.
2. Constant function.
3. one-one function.
4. Onto function.
5. Bijective function.

1. Identity function:

A function 'f(x)' is said to be Identity function if $f(x) = x$.

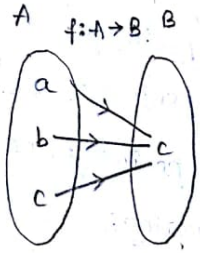
Eg:



2. Constant function:

A function 'f(x)' is said to be constant function if $f(x) = c$, where c is constant.

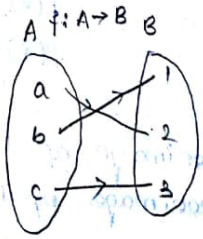
Eg:



3. One-one function:

A function 'f(x)' is said to be one-one function if every element of 'A' must have distinct image in 'B'. (or ea.

Eg:



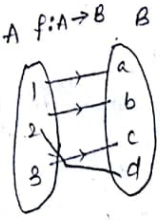
$f(a) = 2$, $f(b) = 1$, $f(c) = 3$.

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4. Onto function:

A function 'f(x)' is said to be onto function if every element of 'B' must have preimage in 'A'.

Eg:



'1' is the preimage of 'a'

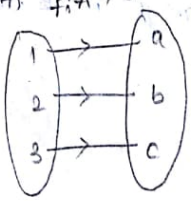
'2' is the preimage of 'b', 'd'

'3' is the preimage of 'c'.

5. Bijective function:

A function 'f(x)' is said to be bijective function if function is one-one and onto.

Eg:



$f(1) = a$

$f(2) = b$

$f(3) = c$.

'1' is the preimage of 'a'.

'2' is the preimage of 'b'.

'3' is the preimage of 'c'.

Example:

1. If $f: N \to N$ and $f(x) = x^2+1$, then check 'f' is bijective or not.

Sol: Given function is $f: N \to N$.

where N is Natural numbers.

$N = \{1, 2, 3, \ldots n\}$.

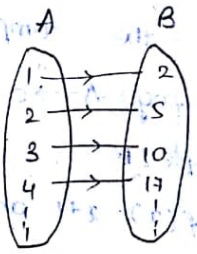
$f(x) = x^2+1$

$f(1) = 1^2+1 = 2$

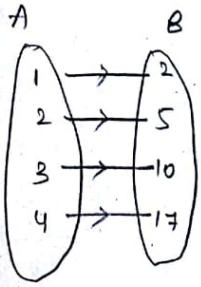
$f(2) = 2^2+1 = 5$

$f(3) = 3^2+1 = 10$

$f(4) = 4^2+1 = 17$.



∴ The above function is one-one function.



'1' is the preimage of '2'

'2' is the preimage of '5'

'3' is the preimage of '10'

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'4' is the preimage of '17'.

∴ the given function is onto function.

∴ it is Bijective function.

11/09/14.

* **Composite Function:**

let us consider $f: A \to B$ and $g: B \to C$, then the composite function is defined as $fog: A \to C$.

Example:

1. let $f(x) = x+1$ and $g(x) = x^2$, then find fog, gof and whether fog = gof or not.

Sol: Given functions

$f(x) = x+1$

$g(x) = x^2$

fog:

$fog(x)$

$= f[g(x)]$

$= f(x^2)$

$= x^2+1 \longrightarrow (1.)$

gof:

$gof(x)$

$= g[f(x)]$

$= g(x+1)$

$= (x+1)^2 = x^2+2x+1 \longrightarrow (2.)$

from (1.) & (2.)

∴ $fog \neq gof$.

2. let $f(x) = x+5$ and $g(x) = x^2+1$, then find fog, gof and whether fog = gof or not.

Sol: Given functions

$f(x) = x+5$

$g(x) = x^2+1$.

fog:

$fog(x)$

$= f[g(x)]$

$= f[x^2+1]$

$= x^2+1+5 = x^2+6$.

gof:

$gof(x)$

$= g[f(x)]$

$= g(x+5)$

$= (x+5)^2+1 = x^2+10x+26 \longrightarrow (2.)$

from (1.) & (2.)

∴ $fog \neq gof$.

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3. let $f(x) = x^2$, $g(x) = x+1$ and $h(x) = x$ then find fogoh, gofoh, gohof and fohog?

Sol. Given functions

$f(x) = x^2$

$g(x) = x+1$

$h(x) = x$

fogoh:

$= fogoh(x)$

$= fog[h(x)]$

$= fog(x)$

$= f[g(x)]$

$= f(x+1)$

$= (x+1)^2 = x^2+2x+1 \longrightarrow (1)$

gofoh:

$= gofoh(x)$

$= gof[h(x)]$

$= gof(x)$

$= g[f(x)]$

$= g[x^2]$

$= x^2+1 \longrightarrow (2)$

gohof:

$= gohof(x)$

$= goh(x^2)$

$= goh(x^2)$

$= g[x^2]$

$= x^2+1 \longrightarrow (3)$

fohog:

$= fohog(x)$

$= foh(x+1)$

$= foh(x+1)$

$= f(x+1)$

$= (x+1)^2$

$= x^2+2x+1 \longrightarrow (4)$

(1) & (4) are equal and

(2) & (3) are equal.

Inverse Function:

A function $f: A \to B$ is said to be invertible, if the inverse relation $f^{-1}: B \to A$ is also function, then f^{-1} is called of inverse of f.

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Eg:

1. let $f(x) = x^2$, then find $f^{-1}(x)$?

Sol: let $f(x) = y$

$x = f^{-1}(y) \longrightarrow (1)$

$\Rightarrow y = f(x) = x^2$

$\Rightarrow y = x^2$

$\Rightarrow x = \sqrt{y}$

from (1)

$f^{-1}(y) = \sqrt{y}$

$\therefore f^{-1}(x) = \sqrt{x}$

2. let $f(x) = x+1$, then find $f^{-1}(x)$?

Sol: let $f(x) = y$

$x = f^{-1}(y) \longrightarrow (1)$

$y = f(x) = x+1$

$y = x+1$

$x = y-1$

from (1)

$f^{-1}(y) = y-1$

$\therefore f^{-1}(x) = x-1$

3. let $f(x) = x+5$ and $g(x) = x^2+1$, then find $f^{-1} \circ g^{-1}$ and $g^{-1} \circ f^{-1}$.

$f(x) = y$

$x = f^{-1}(y)$

$y = f(x) = x+5$

$y = x+5$

$x = y-5$

$f^{-1}(y) = y-5$

$f^{-1}(x) = x-5$

$g(x) = y$

$x = g^{-1}(y)$

$y = g(x) = x^2+1$

$y = x^2+1$

$x^2 = y-1$

$x = \sqrt{y-1}$

$g^{-1}(y) = \sqrt{y-1}$

$g^{-1}(x) = \sqrt{x-1}$

$f^{-1} \circ g^{-1} = f^{-1} \circ g^{-1}(x)$

$= f^{-1}(\sqrt{x-1})$

$= \sqrt{x-1} - 5$

$g^{-1} \circ f^{-1} = g^{-1} \circ f^{-1}(x)$

$= g^{-1}(x-5)$

$= \sqrt{x-5-1} = \sqrt{x-6}$

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4. let $f(x) = x^3$, $g(x) = x^2+1$, and $h(x) = x+1$. find $f^{-1}\circ g^{-1}\circ h^{-1}$, $h^{-1}\circ g^{-1}\circ f^{-1}$ and $g^{-1}\circ h^{-1}\circ f^{-1}$.

Sol.

$f(x) = y$

$x = f^{-1}(y)$

$y = f(x) = x^3$

$y = x^3$

$x = y^{1/3}$.

$f^{-1}(y) = y^{1/3}$

$f^{-1}(x) = x^{1/3}$

$g(x) = y$

$x = g^{-1}(y)$

$y = g(x) = x^2+1$

$y = x^2+1$

$x^2 = y-1$

$x = \sqrt{y-1}$

$g^{-1}(y) = \sqrt{y-1}$

$g^{-1}(x) = \sqrt{y-1}$.

$h(x) = y$

$x = h^{-1}(y)$

$y = h(x) = x+1$

$y = x+1$

$x = y-1$

$h^{-1}(y) = y-1$

$h^{-1}(x) = x-1$

$f^{-1}\circ g^{-1}\circ h^{-1} = f^{-1}\circ g^{-1}\circ h^{-1}(x)$

$= f^{-1}\circ g^{-1}(x-1)$

$= f^{-1}(\sqrt{x-1-1})$

$= f^{-1}(\sqrt{x-2})$

$= (\sqrt{x-2})^3$.

$h^{-1}\circ g^{-1}\circ f^{-1} = h^{-1}\circ g^{-1}\circ f^{-1}(x)$

$= h^{-1}\circ g^{-1}(x^{1/3})$

$= h^{-1}(\sqrt{x^{1/3}-1})$

$= \sqrt{x^{1/3}-1}-1$

$g^{-1}\circ h^{-1}\circ f^{-1} = g^{-1}\circ h^{-1}\circ f^{-1}(x)$

$= g^{-1}\circ h^{-1}(x^{1/3})$

$= g^{-1}(x^{1/3}-1)$

$= \sqrt{x^{1/3}-1-1}$.

$= \sqrt{x^{1/3}-2}$.

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